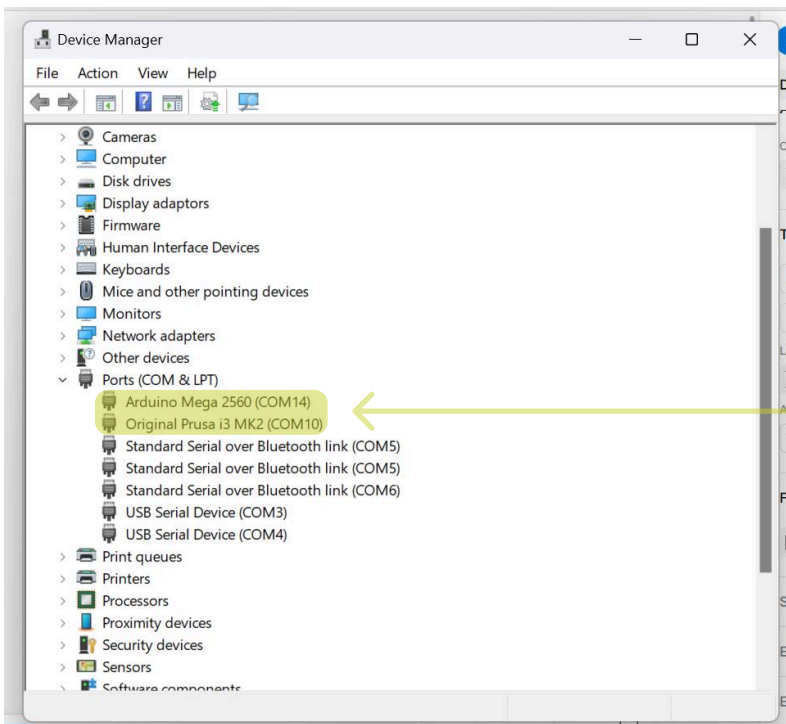


Triboelectric Characterisation Rig: User Guide

Last updated: 2/10/2025

EQUIPMENT SET UP

1. Lift the print head all the way to the top of the z-axis
2. Plug in the 3D printer to mains power
3. Turn the printer on via the power switch
4. Plug in the grey cable (Arduino) and black cable (printer) to your device
5. Open device manager and check which ports they are connected to



EG

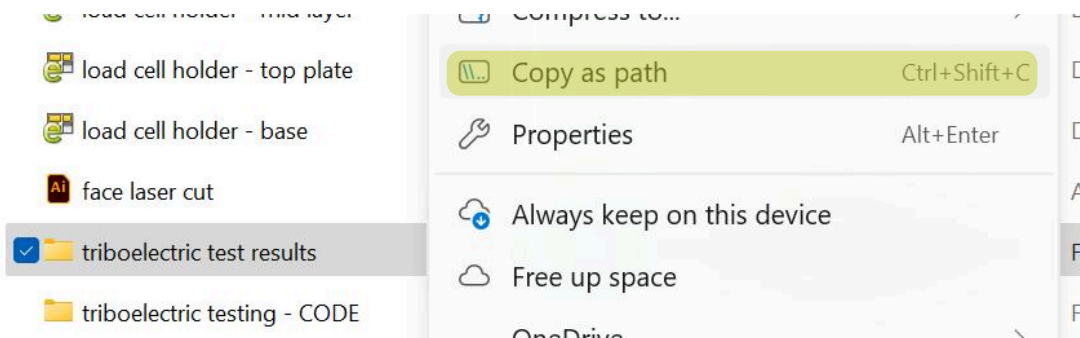
The Arduino is connected via **COM14**
and the 3d printer via **COM10**

SOFTWARE SET UP

1. Open the “Triboelectric_test_programme.py” file in VS Code
2. At the start of the file, change the ‘**arduino_port**’ and ‘**printer_port**’ variables to the correct COM ports for your device

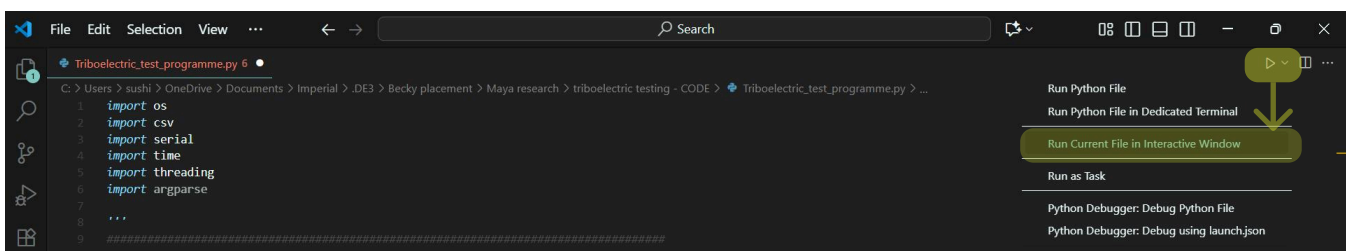
```
23
24 # === UPDATE FOR YOUR SYSTEM ===
25
26 arduino_port = "COM14" # change to the serial port of Arduino
27
28 printer_port = "COM10" # change to the serial port of 3d printer
29
```

3. Change the 'SAV_DIR' variable to where you want the force data files to be saved on your laptop. Make sure to copy the whole file address.

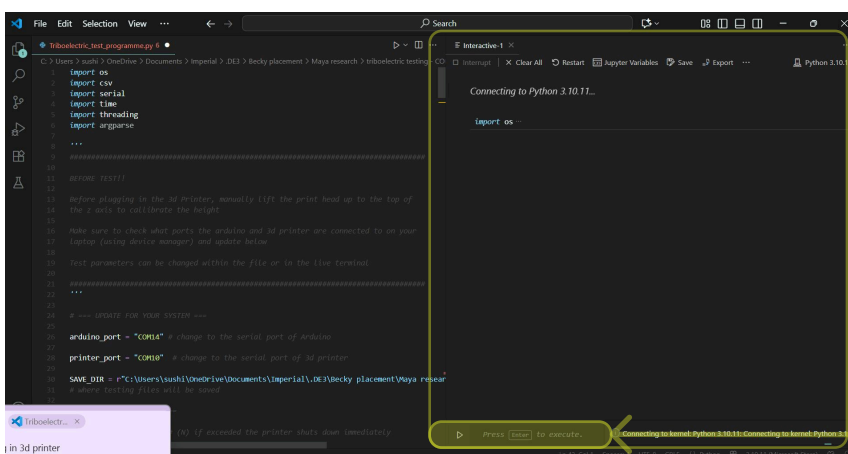


```
SAVE_DIR = r"C:\Users\sushi\OneDrive\Documents\Imperial\DE3\Becky placement\Maya research\triboelectric test results"
# where testing files will be saved
```

4. Run the file in an interactive window (drop down the menu by the triangle play button, then elect "run current file in interactive window"

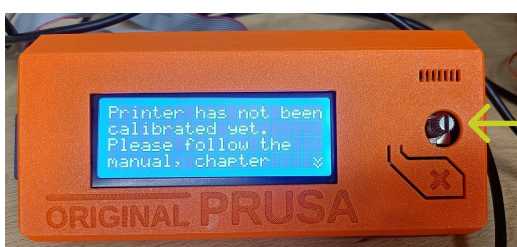


5. A new window should show up with the interactive terminal



this is where you write the
commands you want to send

6. The orange PRUSA screen should show a message about calibration, ignore this and press in the metal button



This button! it doesnt really look like a button
pushing down on the top will give a solid click

The equipment should now be ready for use, and set up to be controlled by the Python file.

Below are the instructions for interacting with the testing code:

MANUAL SET UP AND CALIBRATION

1. Double check the print head has been lifted all the way to the top of the z-axis
 - a. IF NOT → turn off the 3D printer, lift the axis up, then turn the printer back on again
2. run the “**setup()**” function via the interactive window by typing it in to the text box at the bottom. This will:
 - a. calibrate x,y and z axis
 - b. set up the coordinate system
 - c. navigate to the centre point of the test zone in x-y
 - d. navigate to z = 85 (the standard ‘neutral’ height)

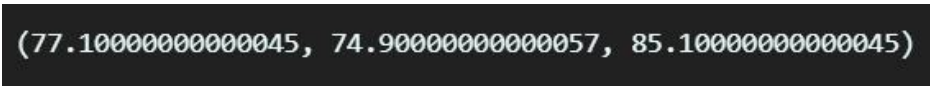


```
▷ setup()
```

3. run “**calibrate_z()**” function. This will:
 - a. slowly step down until the “**threshold_force**” and “**contact_force**” are reached
 - b. return the z coordinate values for “**neutral_z_coord**”, “**contact_z_coord**”, and “**separation_z_coord**”



```
▷ calibrate_z()
```



```
(77.10000000000045, 74.90000000000057, 85.10000000000045)
```

RUNNING ONE OFF CYCLES

1. To run one contact or sliding test cycle, call either of these functions: “**contact_cycle()**” or “**sliding_cycle()**”
2. Both will run a complete set according to the parameters listed at the top of the code. These can be temporarily changed by typing “[variable_name] = [new value]” into the interactive window which will update this value for all future tests run in that window.
3. To update a value permanently, change the variable value within the main python code. To see that update in the interactive window, you need to close the current interactive window, save your python file, then reopen the interactive window.

RUNNING A TEST AND RECORDING FORCE DATA

1. To run only one testing cycle for either contact or sliding, call the "**start_test(a,b)**" function.

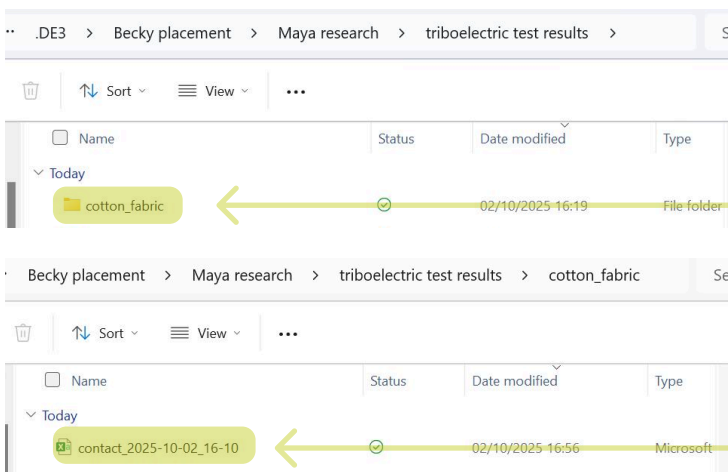
You need to put in parameters a and b where:

a = testing type (you can either have 'contact' or 'slide')

b = material being tested (write text within "" here, using "_" instead of spaces)

```
▶ start_test('contact', "cotton_fabric")
```

2. This will automatically run the full cycle (according to the testing parameters) and record the force data to CSV. A new folder with the name of the material will be created in the testing folder you set as the '**SAV_DIR**' variable. The CSV data will be saved within the material folder with the file name showing the testing mode, and date of the test.



The material folder within the "triboelectric test results" folder

The CSV data for the contact testing cycle, with date formatted as "YYYY-MM-DD_HH-MM"

Once the testing parameters are fully defined and the large scale testing is ready to occur, the automatic testing functions can be used

RUNNING AUTOMATIC TESTS

1. "run_contact_protocol(b)" where b is the material name (as above) will run the full setup and calibration, then complete how ever many rounds of the contact cycle testing as defined by "number_of_contact_tests"
2. "run_slide_protocol(b)" is the exact same as above, but for sliding and completes as many testing rounds as defined by "number_of_slide_tests"
3. "run_test_protocol(b)" will run setup, callibration, and both contact and sliding tests

For even greater control, all parameters can be modified and other functions can be called within the interactive window.

TEST PARAMETERS

These can all be changed temporarily within the interactive window, or permanently in the Python file

```
34 # === TEST PARAMETERS ===
35
36 safe_force_limit = 20 # (N) if exceeded the printer shuts down immediately
37
38 touch_threshold = 0.5 # baseline force (N) that indicates initial contact
39
40 contact_force = 10 # force delivered (N)
41
42 contact_time = 1 #length of time the electrodes are held together (s)
43
44 separation_time = 1 #length of time the electrodes are held apart (s)
45
46 separation_height = 8 # maximum distance between electrodes (mm)
47
48 no_contact_cycles = 2 # number of contact test iterations
49
50 no_slide_cycles = 2 # number of sliding test iterations
51
52 contact_speed = 500 # how fast the downwards load is applied during contact separation (mm/s)
53
54 slide_speed = 1000 # how fast the lateral sliding occurs (mm/s)
55
56 number_of_contact_tests = 2 # number of times the whole contact test is run
57
58 number_of_slide_tests = 1 # number of times the whole sliding test is run
59
```

CORE PRINTER FUNCTIONS

▷ `z_go_to(height)`

→ Makes the printer move to the coordinate 'height'. Assuming setup has been run, the bottom of the 3D printer is z=0, and the top is z=200. **To avoid collisions, never set z under 80**

▷ `x_y_centre()`

→ Moves the printer to the centre of the testing zone

▷ `setup()`

→ Calibrates x, y and z axis

→ Moves the printer to the x_y centre point, and to z=85

▷ `calibrate_z()`

→ Finds the coordinates where threshold force and contact force are reached

→ Calculates and returns separation coordinate based on the set separation distance parameter

▷ `reset()`

→ Moves the printer to the x_y centre and the top of the z axis, z=200

→ Leaves the printer in the correct position to be turned off

BASIC TRIBO TEST MOVEMENT FUNCTIONS

▷ `contact_cycle()`

Runs a full contact cycle:

1. push down to deliver '`contact_force`'
2. hold for '`contact_time`' seconds
3. lift up to separated height
4. hold for '`separation_time`' seconds
5. Repeat steps 1-4 for as many times as '`no_contact_cycles`' says

▷ `sliding_cycle()`

Runs a full sliding cycle:

1. push down to deliver '`contact_force`'
2. Slides across the fabric in the pattern:
 - a. Edge one
 - b. Centre
 - c. Edge two
 - d. Centre
3. Repeats the sliding pattern for as many cycles as '`no_slide_cycles`' says
4. Lifts up to separated height

TESTING FUNCTIONS FOR DATA COLLECTION

`start_test(test_name, material)`

- Runs either contact or slide test
- Saves CSV file in material folder

`run_contact_protocol(material)`

- Runs setup and calibration
- Runs X rounds of contact testing as defined by '`number_of_contact_tests`'
- Saves all CSV files in material folder

`run_slide_protocol(material)`

- Runs setup and calibration
- Runs X rounds of slide testing as defined by '`number_of_slide_tests`'
- Saves all CSV files in material folder

`run_test_protocol(material)`

- Runs setup and calibration
- Runs X rounds of contact testing as defined by '`number_of_contact_tests`', followed by X rounds of slide testing as defined by '`number_of_slide_tests`'
- Saves all CSV files in material folder